

1. Rationale and scope

Chosen field type: intermediate-frequency alternating magnetic field (AMF/LF-EMF), sinusoidal, 100-320 kHz. This is selected over nsPEF, ELF-EMF, SMF, RF-EMF, PMF, MMW/THz and TTFields because it offers the best combination of a magnetic-only delivery rationale, a defined anti-tumoral parameter window, and direct cancer-vs-healthy viability selectivity in the report. The strongest AMF evidence is Paper 4: 227 kHz/18 mT/30 min reduced viability across 9 cancer lines (48-88% of control) while sparing NHA, HCF and HUVEC (90-105% of control); the broader AMF category contains 38 viability conditions, with cancer reductions in 15/27 and no viability loss in 11/11 healthy measurements. Report pp. 22-24.

Why not TTFields: excluded as already FDA-approved. Why not MMW/THz: the report identifies the cleanest single matched selectivity signal in H1299 vs MCF-10A, but upstream mechanism is unresolved because ROS, mitochondrial membrane potential and calcium were not measured. Why not ELF-EMF: large evidence base but insufficient population-level selectivity and hormetic or healthy-cell toxicity counterexamples. Why not nsPEF/nsPMF/RF/SMF/PMF: either sparse healthy controls, limited paper base, broad parameter heterogeneity, or pro-proliferative windows. Report pp. 2-3, 17-18, 48-49.

Boundary: the hypothesis addresses short-duration, non-thermal, sinusoidal AMF-kHz exposure as monotherapy in proliferative malignant cells, especially glioblastoma/glioma and triple-negative breast cancer. It does not claim a universal EMF mechanism, does not explain TTFields-class electric-field mitotic disruption, and does not infer in-vivo efficacy. It is designed to test the report's unresolved link between AMF-induced redox/antioxidant disturbance and selective loss of cancer-cell viability.

2. Mechanistic hypothesis

Hypothesis statement. A 200-230 kHz sinusoidal magnetic field at 10-18 mT imposes sub-electroporative induced-current stress that is buffered by non-malignant cells but selectively overwhelms cancer cells with vulnerable mitochondrial/redox homeostasis, producing a non-apoptotic metabolic arrest phenotype and delayed viability loss. The expected cell fate is loss of clonogenic growth and viability at 24-120 h, with apoptosis not serving as the dominant early route.

Step 1 - field-cell coupling. Supported in the report: AMF exposures in Paper 4 used sinusoidal 227 kHz/18 mT for 30 min in an actively temperature-controlled non-thermal setup; Paper 4 also screened 160, 198, 227, 280 and 320 kHz and observed maximal sensitivity at 198-227 kHz with reduced effect at 320 kHz. Inferred link: in this frequency-amplitude window, induced cytoplasmic currents, rather than heating or electroporation, are the proximal physical perturbation. Report pp. 22-23, 42-43.

Step 2 - mitochondrial/redox loading. Supported: AMF mechanistic measurements showed ROS Direction = +1 in 2/3 cancer conditions; antioxidant Direction was measured in 9 conditions and was dominantly -1 in cancer cells; the AMF molecular profile included glycolysis enzymes, mitochondrial-biogenesis markers and cell-cycle regulators. Limitation: mitochondrial membrane potential was measured only once in AMF and therefore cannot yet be treated as a demonstrated primary lesion. Inferred link: redox/antioxidant collapse is downstream of mitochondrial electrical or metabolic loading. Report pp. 23-24.

Step 3 - metabolic checkpoint failure. Supported: the report connects AMF-kHz exposure with metabolic-pathway changes and cell-cycle markers, and Paper 38 phase-resolved G0/G1 arrest in SH-SY5Y. Inferred link: cancer cells that fail to restore antioxidant balance enter a G0/G1-dominant, proliferation-incompetent state that culminates in reduced viability or clonogenicity. Report pp. 23-24.

Step 4 - selective cell fate. Supported: in AMF, 0/11 healthy viability measurements were Direction = -1, while 15/27 cancer viability measurements were Direction = -1; in Paper 4 the same 227 kHz/18 mT exposure spared NHA, HCF and HUVEC. Inferred link: healthy cells buffer the induced-current/redox perturbation through intact antioxidant reserve and avoid the metabolic arrest cascade. Report pp. 22-24.

Key constraint. The report explicitly states that the strongest AMF selectivity result is anchored to a single paper, that independent replication at 227 kHz/18 mT is absent, and that antioxidant rescue has not yet linked ROS/GSH disturbance causally to viability loss. The proposed design treats these as primary experimental gaps, not settled facts. Report p. 24.

3. Parameter selection strategy

Parameter	Primary test window	Report-derived rationale	Built-in stress test
Frequency	198, 227 kHz; flankers 160, 280, 320 kHz	Paper 4 showed maximal cancer sensitivity at 198-227 kHz and weaker effect at 320 kHz; cross-synthesis places the magnetic sweet spot at 100-300 kHz.	A true window should peak at 198-227 kHz and attenuate at flankers.
Intensity	10, 14, 18 mT; low-dose 1.5 mT	Effects were weak near 1.4-1.6 mT, large at 10-18 mT; 18 mT is the matched Paper 4 selective condition.	Low-dose 1.5 mT should be weak or null; toxicity in healthy cells at 18 mT would reject selectivity.
Waveform	Continuous sinusoid	All AMF exposures in the module were sinusoidal; waveform should be held constant to isolate frequency/intensity.	No waveform mixing in the primary experiment.
Exposure duration	30 min single exposure; 30 min/day x 5; optional 60 min	Paper 4: 30 min; Paper 38: 60 min; Paper 44: 60-90 min. Cross-synthesis proposes 30-60 min daily over 5 days.	Distinguish acute redox response from cumulative loss of proliferative capacity.

4. Experimental design

Cell-line panel. Cancer: U87, LN229, U251, A172 and T98/T98G glioblastoma/glioma lines to reproduce Paper 4's responsive brain-tumor panel; MDA-MB-231 and MDA-MB-453, plus MCF-7 as a cross-line breast benchmark because all were included in AMF Paper 4 and showed reductions at 227 kHz/18 mT. Healthy: NHA, HCF and HUVEC because they were the matched spared controls in

Paper 4; HDFa because it was included in Paper 38; MCF-10A or MCF-12 may be added only as a breast epithelial comparator, with results interpreted as cross-report rather than AMF-matched evidence. Report pp. 22-23.

Exposure apparatus. Use a custom solenoid or Helmholtz-coil exposure system, matching the AMF module devices, with sinusoidal current drive, field mapping across each culture position, sham coils, identical incubator/gas conditions, and active temperature control. Required non-thermal controls: real-time medium temperature in exposed and sham plates; identical thermal history sham; 18 mT static-field-only control to separate time-varying induction from DC magnetic exposure; frequency-off controls at 160/320 kHz; randomized plate positions; blinded outcome acquisition. The report identifies active temperature control and non-thermal operation as core AMF setup features. Report p. 22.

Design. Phase I parameter map: every cell line receives sham, 1.5 mT/227 kHz, 10 mT/227 kHz, 14 mT/227 kHz, 18 mT at 160/198/227/280/320 kHz, single 30-min exposure; outcomes at 0, 6, 24 and 48 h. Phase II cumulative monotherapy: the best selective acute condition plus one submaximal condition, 30 min/day for 5 days, read at days 3, 5 and 7. Healthy controls are run in the same plate format, passage range and exposure session as their cancer comparators.

Core outcomes. Cell fate: viability/metabolic assay, live-cell confluence, EdU/Ki-67 or cell counts for proliferation, clonogenic survival, Annexin V/PI, caspase-3 cleavage, necrosis marker, and cell-cycle phase by PI/FxCycle. Mechanistic: ROS by CellROX or DCFH-DA, GSH/GSSG or equivalent antioxidant pool, mitochondrial membrane potential by JC-1/TMRE at 0/6/24 h, morphology/cytoskeleton imaging, and metabolic markers aligned to LDH-A, HK2, PKM2 and PGC-1 α . Molecular: γ -H2AX foci/comet endpoints, cyclin D1 and p21. These endpoints map directly to the AMF module's measured ROS/antioxidant, metabolic, DNA-integrity and cell-cycle blocks and to its unresolved $\Delta\Psi_m$ gap. Report pp. 23-24, 43.

Selectivity analysis. Define a pre-specified selectivity index as cancer viability reduction divided by healthy viability reduction at identical parameters and observation time; also require no Direction = -1 for healthy viability/proliferation. Because the report warns that healthy-cell inclusion is a major limitation across EMF categories, each frequency-intensity point must be evaluated in all healthy controls before being called selective. Report pp. 8, 43.

5. Predictions and falsification criteria

Predicted positive pattern. The 198-227 kHz/10-18 mT window should produce the strongest effect in glioblastoma/glioma and breast cancer lines, with weaker or absent effects at 1.5 mT and at 160/320 kHz. Sensitive cancer lines should show early ROS increase and antioxidant depletion, followed by altered $\Delta\Psi_m$ if it is the upstream lesion, increased G0/G1 fraction, reduced proliferation/clonogenicity, and delayed viability loss. Apoptosis should be absent or secondary, consistent with AMF apoptosis Direction = 0 in the report. Healthy NHA, HCF, HUVEC and HDFa should remain Direction = 0 for viability/proliferation and should show transient or absent redox changes that normalize by 24 h.

Parameter-specific predictions. A single 30-min 227 kHz/18 mT exposure should reproduce the Paper 4 rank order qualitatively: clear reductions in U87/A172/LN229/U251/MCF-7/MDA-MB-231-class cancer cells, sparing of NHA/HCF/HUVEC. Daily exposure should deepen proliferation/clonogenic effects without converting healthy-cell sparing into toxicity. The 1.5 mT arm should be mostly null except possibly in the most sensitive cancer contexts, matching the weak effects in Paper 38; 10 mT should be intermediate, consistent with Paper 44 reductions in MDA-MB-231 but needing healthy controls.

Falsification. The hypothesis is rejected if (i) 227 kHz/18 mT fails to reduce viability or clonogenicity in the responsive cell lines identified by the report under verified non-thermal conditions; (ii) healthy cells show comparable Direction = -1 viability/proliferation at the same parameter points; (iii) redox/antioxidant and cell-cycle changes do not precede or correlate with loss of viability; (iv) the frequency response is flat rather than windowed; or (v) thermal-matched sham reproduces the effect. It is revised, not confirmed, if apoptosis is the dominant early death route, because that conflicts with the AMF module's uniformly Direction = 0 apoptosis pattern.

Evidence note: all page references refer to the attached report, Electromagnetic fields as monotherapy on cancer cells in vitro: a literature-based dataset analysis / Non-Thermal Electromagnetic Fields as a Selective Antitumor Monotherapy, pp. 1-49.